

BIT4107-Mobile Application Development

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Week 1: Introduction to mobile development.

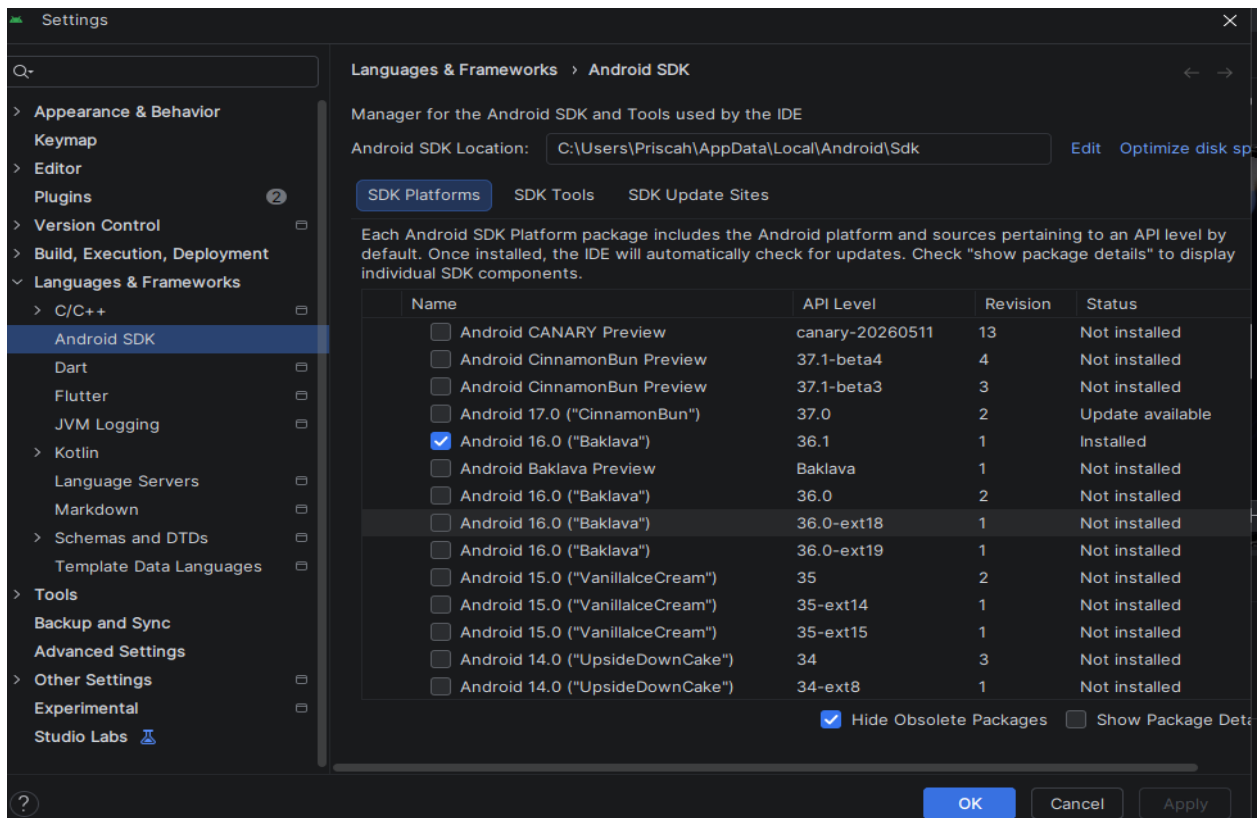


Figure 1.1 Android Studio SDK Manager window showing installed platform tools.

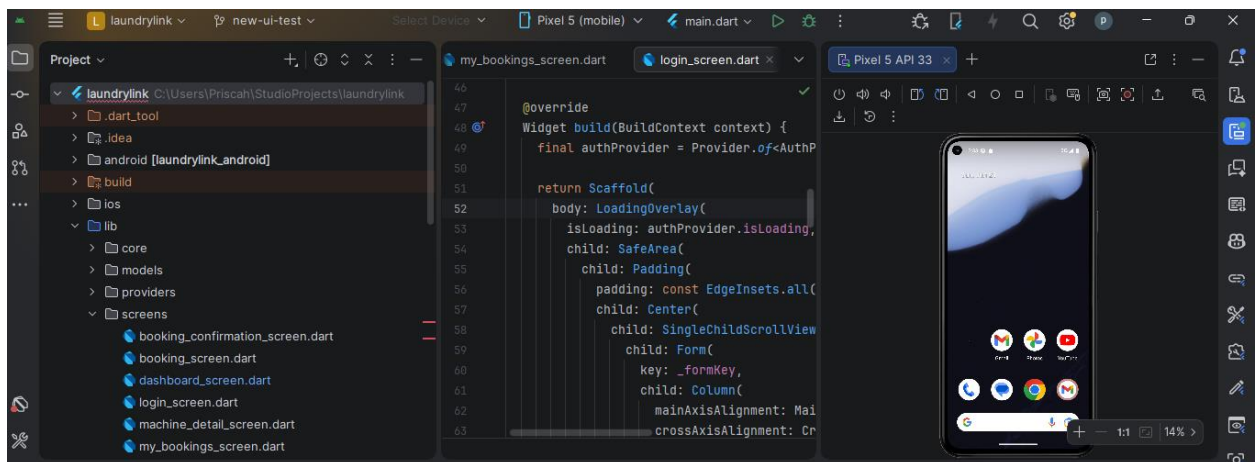


Figure 1.2 Android Virtual Device (AVD) manager interface displaying a newly created emulator profile.

1.1 Tools/software used

1. Android Studio
2. Flutter SDK
3. Android Smartphone
4. Dart Plugin

1.2 Personal reflection

Setting up Android Studio required careful configuration of system paths and the SDK manager. Managing these initial dependencies was challenging but crucial for ensuring that the development environment could compile mobile applications smoothly.

Week 2: Mobile Development Environment.

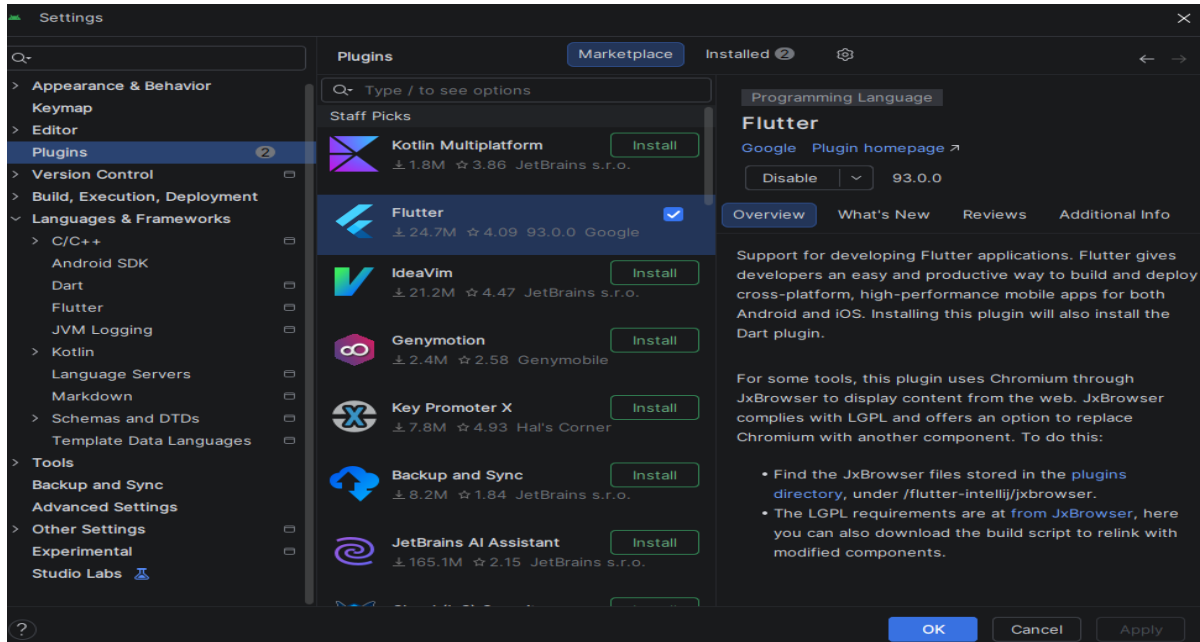


Figure 2.0 Android Studio Plugins Marketplace

```
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Priscah>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[v] Flutter (Channel stable, 3.44.2, on Microsoft Windows [Version 10.0.22000.2538], locale en-CA)
[v] Windows Version (11 Pro 64-bit, 21H2, 2009)
[v] Android toolchain - develop for Android devices (Android SDK version 36.1.0)
[v] Chrome - develop for the web
[X] Visual Studio - develop Windows apps
   X Visual Studio not installed; this is necessary to develop Windows apps.
     Download at https://visualstudio.microsoft.com/downloads/.
     Please install the "Desktop development with C++" workload, including all of its default components
[v] Connected device (3 available)
[v] Network resources

! Doctor found issues in 1 category.

C:\Users\Priscah>
```

Figure 2.1 Terminal window running

2.1 Personal reflection

Integrating the Flutter plugin into Android Studio allowed me to leverage cross-platform tools. Running the doctor command line tool verified that my toolchain was fully compliant and ready for project initialization.

Week 3: User Interface Design



Welcome to LaundryLink



Login

Don't have an account? [Register](#)

Figure 3.0 Login Screen showing the sign-up form and input field layout.

Register











Register

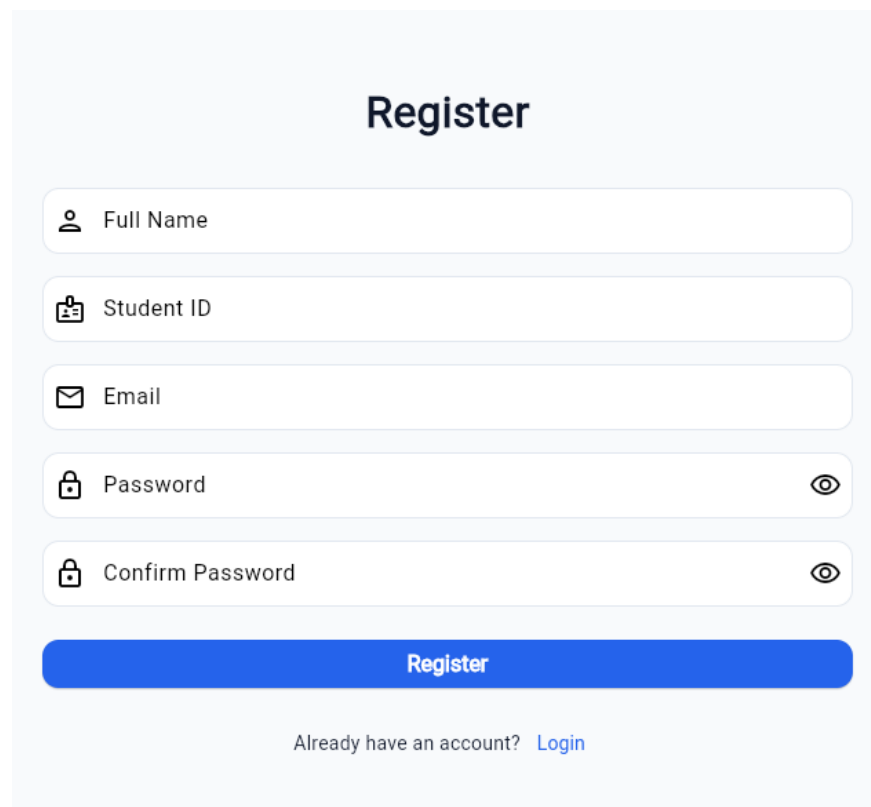
Already have an account? [Login](#)

Figure 3.1 Registration Screen layout showing the main sign-in page design.

3.1 Personal reflection

This week I learned that sketching out the login and signup screens first saved me a lot of time. It helped me figure out where to place the text boxes and buttons so the app feels clean and easy for a student to use right away.

Week 4 : event handling and user interaction



The image shows a registration screen with a light blue background. At the top, the word "Register" is centered in a bold, dark blue font. Below it are five input fields, each with a small icon on the left: a person icon for "Full Name", a student ID card icon for "Student ID", an envelope icon for "Email", a padlock icon for "Password", and another padlock icon for "Confirm Password". The "Password" and "Confirm Password" fields have an eye icon on the right to toggle visibility. Below the input fields is a prominent blue button with the word "Register" in white. At the bottom, the text "Already have an account?" is followed by a blue link labeled "Login".

Figure 4.0 Registration screen



Welcome to LaundryLink

Email

 Sandra@mkuniversity.edu

Password

Invalid email or password

Login

Don't have an account? [Register](#)

Figure 4.1 Validation error

4.1 Personal reflection

This week I learnt about implementing event handling across the registration and login screens. Button tap events triggered form validation, giving users instant feedback when fields were left empty or passwords did not match. The password visibility toggle demonstrated how a single user interaction can change the state of the entire screen dynamical.

Week 5: Data Management

 **Book Machine**

Select Machine

Washer 1 

Select Date

Wed, Jun 17, 2026 

Select Time

8:00 AM	8:30 AM	9:00 AM	9:30 AM	10:00 AM	10:30 AM	
11:00 AM	11:30 AM	12:00 PM	12:30 PM	1:00 PM	1:30 PM	
2:00 PM	2:30 PM	3:00 PM	3:30 PM	4:00 PM	4:30 PM	5:00 PM
5:30 PM	6:00 PM	6:30 PM	7:00 PM	7:30 PM		

Confirm Booking

Figure 5.0 Booking screen



Booking Confirmed!

Your booking has been successfully made

Washer 1

Confirmed

📅 Wed, Jun 17, 2026

🕒 9:30 AM - 10:00 AM

📍 Ground Floor

[View My Bookings](#)

[Back to Home](#)

Figure 5.1 Booking Confirmation

← My Bookings

Dryer 6

Confirmed

📍 First Floor

📅 Mon, Jun 29, 2026

🕒 9:00 AM - 9:30 AM

⊗ Cancel Booking

Washer 5

Confirmed

📍 First Floor

📅 Fri, Jun 26, 2026

🕒 10:00 AM - 10:30 AM

⊗ Cancel Booking

Washer 1

Confirmed



Home



My Bookings



Profile

Figure 5.2 Bookings list

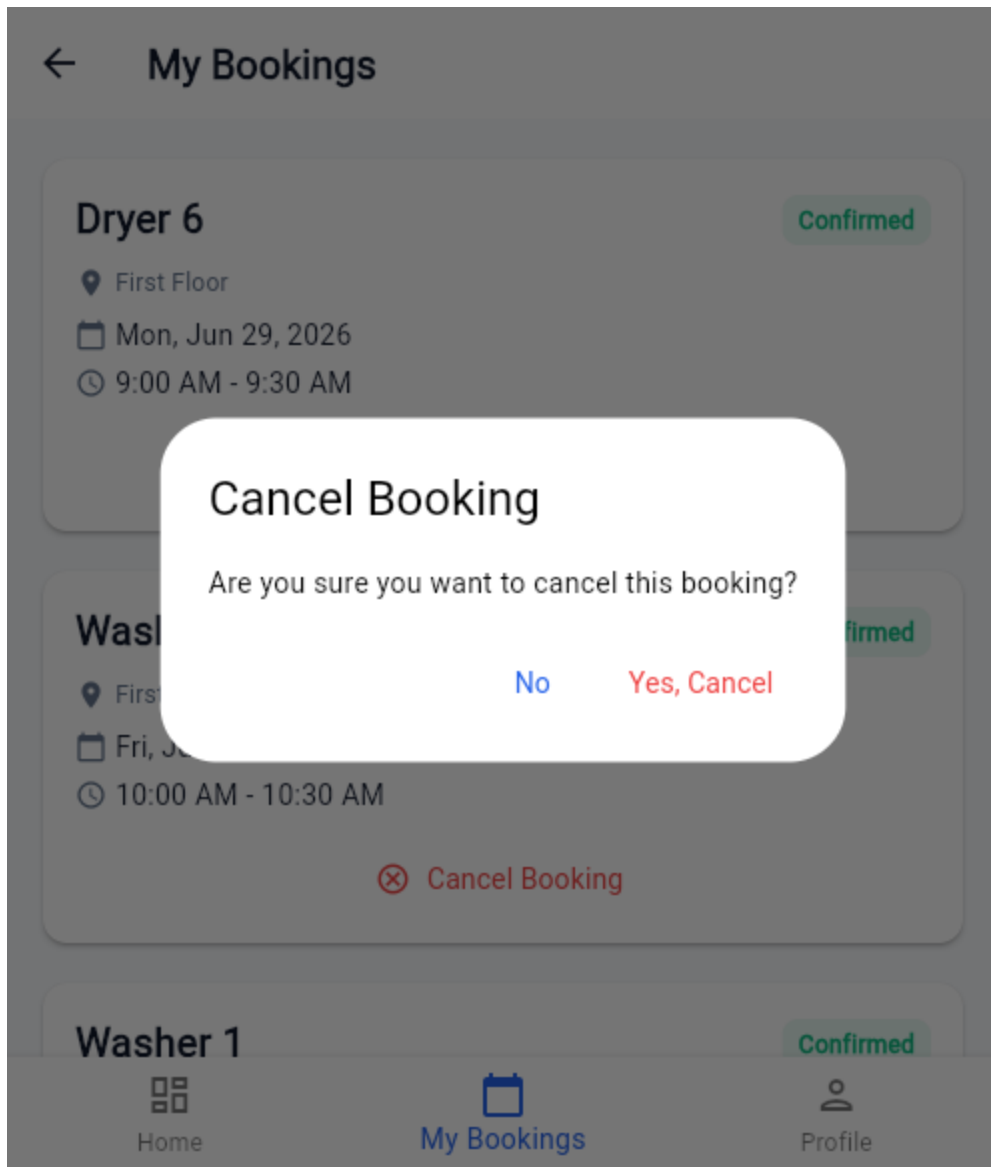


Figure 5.3 Cancel bookings

5.1 Personal Reflection

This week I applied CRUD operations within the LaundryLink app. Creating a booking stored the student's reservation, Reading retrieved available machines and existing bookings, Updating allowed modification of booking details, and Deleting removed unwanted reservations. This demonstrated how structured data management directly improves the reliability and usability of a mobile application..

Week 6 : Networking and APIs

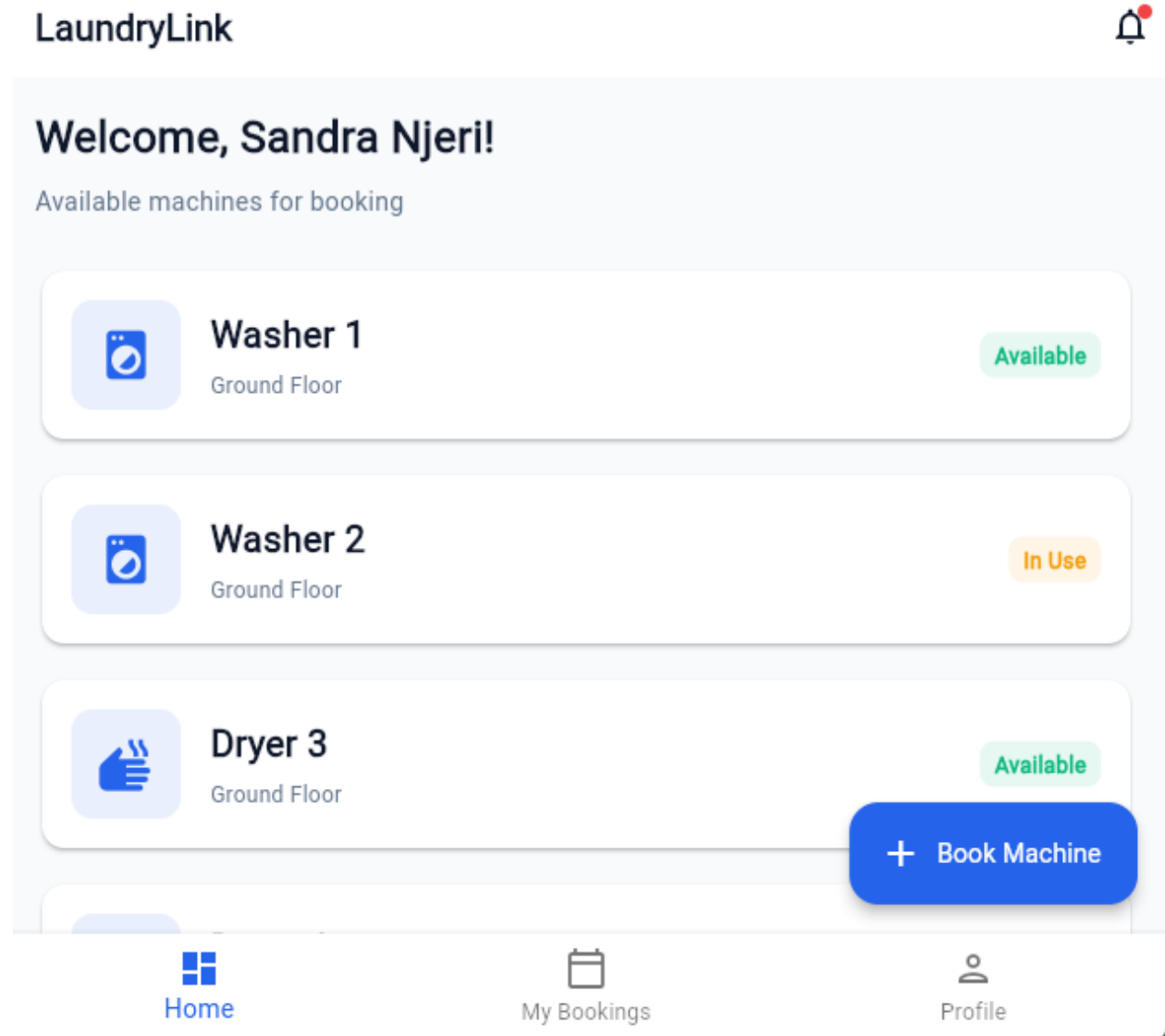


Figure 6.0 Machine Detail Screen showing live machine availability.

← Booking Confirmed



Booking Confirmed!

Your booking has been successfully made

Washer 1

Confirmed

📅 Wed, Jun 17, 2026

🕒 9:30 AM - 10:00 AM

📍 Ground Floor

[View My Bookings](#)

[Back to Home](#)

Figure 6.1 My Bookings- showing the student's saved reservation list.

6.1 Personal Reflection

This week, integrating REST APIs with async/await requests allowed LaundryLink to fetch live machine statuses via GET and sync bookings via POST. This taught me how to handle asynchronous data flows and to emphasize local caching to prevent data loss.